Origins of Dragons: A Literature Review

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Abstract

Dragons are a feature in the myths of many human civilizations, whether it be the European dragon of the Middle Ages, the auspicious Chinese loong, the aggressive Mediterranean sea serpent, or the feathered serpent of Mesoamerican folklore. The common presence of such beings in myths, despite differences in geography, language, religion, and thousands of years of independent development, remains a debated mystery in the realm of comparative mythology and anthropology. This literature review will attempt to analyze the contributions of the four major frameworks for studying dragon mythology in their ability to account for this phenomenon in its entirety. The cognitive framework, first posited by Jones (2002) in his theory of the "predator template" and further confirmed by Kawai and He's (2016) research on human snake detection abilities, holds that dragons are cognitive universals resulting from evolved predator threat-detection systems. The ecological framework considers the influence of local fauna and landscape, as well as fossil discoveries, on the particular morphological characteristics in draconic imagery. The symbolic framework describes the use of dragon imagery as tools within local ideological, theological, and cosmological systems, including its function as a symbolic representation of imperial rule in China (Zhang 2024) and of sin and chaos in medieval Christian Europe (Khalifa-Gueta 2001). Finally, the cultural diffusionist framework examines how draconic symbolism is historically transmitted via trade routes and other channels of cultural communication, while also incorporating an explanation based on communicative translation practices (Luo 2022) that result in cross-cultural similarity. Upon examination, this paper suggests that a monodisciplinary approach to the subject is insufficient and that the phenomenon of cross-culturally shared dragon images should be considered through an integration of all four frameworks. The evolutionary psychology framework identifies the cognitive prerequisites; the ecological perspective explains their manifestation in local cultural terms; the symbolic framework accounts for the process of cultural elaboration and importance; cultural diffusion produces other similar features as a result of contact, while including its linguistic dimension of translation, and how so many morphologically varied constructions become defined as the same creature family. Overall, the value of the discussed model lies in its applicability to other cases of mythological universals.

Introduction

Dragons are among one of the most widespread, varied, and prevalent mythological creatures across the globe. For example, the fire-breathing dragons of Western European medieval folklore stand in stark contrast to the imperial long of ancient China, or the feathered Quetzalcoatl of pre-Columbian mythology differs entirely from the rain-bringing dragons of Japan. It seems that almost every major civilization has its own version of a large, powerful reptilian creature in their mythology. What makes this phenomenon particularly fascinating is its apparent universality and the possible circumstances surrounding its development. Cultures separated by distance, belief systems, differing environments, and even centuries all still have produced myths about these creatures that all seem to possess morphologically and symbolically similar characteristics.

Thus, the two questions I am using to guide my research are: What is the explanation for the cross-cultural appearance of dragon mythology? What do these explanations say about the human relationship to how these people perceive the world around them? Because, according to Zhang (2024), academics generally agree that ‘dragons’ as a word does not encompass one concept, as these creatures have physical and cultural manifestations that vary significantly. As such, the cross-cultural prevalence of dragons is best understood not as the result of a single origin but as the intersection of evolutionary cognition, environmental influences, symbolic systems, and historical interactions, demonstrating the need for a multi-framework anthropological approach to myth analysis. As well, dragons provide a case study into how the human imagination has a tendency to fill in gaps in knowledge to better explain the unknown.

Methodology

As this is a literature review, there was no need for an IRB certification or formal ethics review. I started out by identifying four different ‘dragon types’ to compare and contrast. I chose Western European and East Asian dragons due to their prominence in global knowledge, Western American cultural knowledge shaped by Hollywood, and the fact that these are typically what come to mind when someone hears the word ‘dragon’. Additionally, I chose to analyze the myths of Mediterranean sea serpents for their portrayal of large serpentine creatures within the Mediterranean region, as they are often included in the morphological family of dragons. Finally, I chose to analyze the Mesoamerican myths of amphitheaters for their representation in the Americas and to demonstrate that dragons are worldwide, not exclusive to the Old World.

I collected articles from popular, accredited, and respected academic journals, as well as online and physical books, for my sources. While collecting my research, I noticed a vast disparity in the cultural representation of dragon research: most focused on Western European and East Asian dragons, with very little on Mesoamerican dragons, and Mediterranean dragons in the middle. This can be, in part, attributed to the cultural prominence of Western European and East Asian dragons in worldwide knowledge due in part, to these regions' history of colonization, thus spreading their myths to whom they colonized, which then grew them to the prevalence they have today in modern television and cinima; however, there is also a disparity in the classification of 'dragon' in academic literature.

Culturally, Amphithere is a European term for creatures with serpentine bodies, occasionally a beak, and no limbs other than wings (Rose 2001). Oftentimes, these wings can be either feathered or more leathery, like bat wings. The term was applied to certain Mesoamerican gods, such as the Maya god Quetzalcōātl, due to their resemblance to the European creature. Later, these depictions and myths became better known as amphitheaters, overshadowing the original European myths in popular knowledge (Rose 2002). Though I will be using the term in this paper to refer to the Mesoamerican dragons, as they all are similar in appearance due to cultural diffusion in the area, it is important to recognize how the term is inherently colonial and is ascribed to creatures that are typically deities, as compared to the more animalistic nature described in draconic myths in the Old World.

Cognitive Framework

The book, *An Instinct for Dragons*, David E. Jones (2002), theorizes that dragons must be interpreted as composite images that can be termed as a "predator template," made up of elements from the three most threatening kinds of predators during the times when our ancestors had to survive against these animals: the big cats and raptors, which were capable of attacking humans from above; snakes, which attacked from below and could paralyze or even kill its victim with venom; and other big primates that threatened our early ancestors socially. Jones argues that, through millions of years, our human ancestors' brains were able to evolve while maintaining their regard to how they responded to certain kinds of threatening stimuli from these animals. What then resulted was an innate system within our human brains which recognized danger based on very distinct visual and behavioral characteristics of these animals, which, when

triggered in cultural contexts would cause the human mind to create images like those of dragons: big body frames, scales and other reptilian-like characteristics, claws, fangs, sometimes wings (Jones 2002).

Academic support for Jones' hypothesis, at least in regard to the reptilian portion of the "predator template", can be supported by a 2016 study in cognitive science. In their study, Kawai and He (2016) demonstrated that humans detect snakes statistically faster and with greater accuracy than other animals, even under obscured conditions. This then suggests that snake detection still remains a basic skill within human cognition, even after millennia of evolution. Kawai and He's findings are consistent with the broader academic writings on what is called the "snake detection hypothesis." This hypothesis proposes that primate visual systems were, in part, shaped by the need to identify camouflaged snakes in forested environments. The cultural prevalence of snake-like imagery in religious iconographies and mythologies, with the tendency for snakes to be depicted as malicious, may in part, demonstrate the deep-seated awareness of snake-like animals. In his article, "The meaning of snakes and dragons across world cultures on religious buildings in their iconography," Hoven (2019) documents the cross-cultural frequency of snake and dragon imagery on religious buildings, spanning cultures as geographically and historically distant as ancient Mesopotamia, medieval Europe, and pre-Columbian Mesoamerica. This distribution is difficult to explain through cultural diffusion alone by how chronologically and geographically separated they were, which then, in turn serves as an argument in favor Jones's predator template hypothesis.

In their article, "Imaginary Worlds Through The Evolutionary Lens: Ultimate Functions, Proximate Mechanisms, Cultural Distribution," Doubourg and Baumard (2022) contextualize this type of reasoning within evolutionary theory, in a way that attempts to explain how the content of fictional stories is shaped by adaptive adversities within human evolutionary history. According to their theory, the emergence of similar dragon-like creatures across various mythologies is not random but is a result of the human ability to solve adaptive problems, like dealing with predators, hierarchies, competition for mates, and unpredictable environments through the use of psychological mechanisms. The presence of similar mythical creatures across cultures is one example of how the human imagination tends to revolve around certain topics and characters because they trigger a particular evolved psychological system.

Despite its merits, the cognitive-evolutionary framework faces several major drawbacks. First of all, it fails to satisfactorily explain the vast diversity of dragons, their mythic archetypes, and their meanings in various cultural traditions. Based on the premises of the predator template model, draconic creatures should always be seen as hostile forces to humans. However, the presence of the benevolent Chinese loong dragons proves that evolutionarily driven inclinations alone cannot fully explain the development of specific forms and symbolic associations associated with dragons, since these factors are largely determined by socio-cultural, historical, and ecological determinants. Where the cognitive framework explains the phenomenon of the dragon myth, it alone cannot explain the prevalence of dragon-like creatures across the world.

The Ecological Approach

The Ecological framework centers on the role of environmental influences in creating myths regarding dragons. The key idea behind this framework is that people understand their surroundings in terms of what they see and experience, and before the rise of zoology and paleontology as disciplines, there were no explanations for unusual animals, fossils, dramatic natural scenery, or unexplainable natural phenomena other than the dragon myth. The key idea behind this framework is that people understand their surroundings in terms of what they see and experience, and before the rise of zoology and paleontology as disciplines, there were no explanations for unusual animals, dramatic natural scenery, or unexplainable occurrences other than the dragon myth.

The paleontological argument has perhaps the strongest supporting evidence behind it. In the past, a finding of massive unknown bones or skulls in some geological formation, without the knowledge that modern science has given us, would have caused them to be seen as relics of strange and previously unknown creatures, rather than the prehistoric dinosaurs we know them to be. According to Michael Lorenzen (2025), in his writing, “Dragons in Mythology: A Literature Review,” while examining literature concerning this issue, many dragon myths found in various cultures seem to be concentrated in locations where there is a particularly dense presence of easily-accessible dinosaur or reptilian fossils. A good case-study in this point is from what is geologically now modern-day China. A fossil, dubbed *Dinocephalosaurus orientalis*, was discovered off the coast of southwestern China, Spikeman et al. (2024). The creature is what is known as an “aquatic archosauromorph,” dating to the Middle Triassic. It has unique anatomical

features, including an extremely long neck, a serpentine body, and small fore and hind legs. This morphology is similar to the traditional imagery depicting Chinese dragons. Although a direct cause-and-effect relationship cannot be identified, it is interesting enough that fossils of such type, and with such a strong resemblance to mythological depictions in that area, coincide with a strong tradition of dragon mythology. The same reasoning is provided by Rush (2021), in *The Origin of the Dragon,* which suggests that seeing fossilized bones of large and unfamiliar creatures could lead to powerful imaginations in people, and the dragon myth serves as a means to explain and classify them.

The ecological framework not only refers to fossils, but also to living animals. As Jones (2002) notes, there are several instances where a natural explanation for some reports about encounters with sea serpents or other monstrous beings is plausible. One such instance is a hypothesis that reports of Mediterranean sea serpents could have been caused by people witnessing the erect penises of massive whales when the creatures stick them up from the water during mating displays. Such an image would certainly be enough to give witnesses, who are unfamiliar with whale biology, the impression of having seen some kind of monster in the water. As pointed out by Núñez et al.(2015) in their article, "The Serpent as a Pan-Mediterranean Myth," the sea serpent tradition of the Mediterranean area existed not because of mythology alone, but also because of the dangerous and unpredictable nature of the waters themselves, characterized by terrible storms and bizarre marine life.

The ecological argument has another instance in Mesoamerica. The combination of the morphology of a snake and a bird in the feathered serpentine creatures dubbed amphitheres could be interpreted as a reflection of the cultural interpretation of actual meetings with unique or intimidating animals in the dense forest environment of Mesoamerica. According to Jones (2002), the winged or feathered reptilian in Mesoamerica may represent the disorienting sight of a flock of birds flying across tree tops, from certain angles appearing like the body of a twisting creature. The link between the feathered serpent and wind and agricultural fertility is also evident in its mythic significance, indicating that the myth partially acted as an explanation for natural phenomena.

The ecological approach works when applied in conjunction with the cognitive approach. This is because natural selection and evolution make some stimuli particularly noticeable and

memorable, such as large bones and strange creatures, whereas local ecology specifies what particular stimuli will become available for perception by people in their environment.

Symbolic Approach

The third framework focuses on viewing dragon myths' importance not from the perspective of psychology or from the ecological perspective, but from the perspective of their role in culture. This approach is based on considering them as symbolic structures that derive their meaning from connection to other symbols, as well as to the various issues of the societies that generate them. From this point of view, the importance lies more in exploring the use of dragon myths after they emerge, and the importance they garnered within their culture of origin which would allow them to spread, rather than in seeking an explanation for their emergence.

Perhaps the clearest example of symbolic variation is the contrast between the representation of dragons in Western European and East Asian cultures. In medieval Western Europe, dragons were considered almost an embodiment of all things sinful and chaotic - a monster representative of Satan and all things that were perceived as threats to the theological and social order through heresy, violence, and moral corruption. Khalifa-Gueta (2001) argues that this symbolism has its roots in classical Mediterranean myths, where it evolved into Christian imagery representing the triumph of virtuous saints over evil, represented by the serpent myth of Roman antiquity, such as in the story of Saint George. Not only did this symbolic function serve an artistic purpose but was also very actively used to establish the legitimacy of Christian ideology, crusades, and as a moral warning for going against the teachings of the religion.

In contrast, however, in East Asia, their dragon depictions have the opposite symbolic meaning. As argued by Zhang (2024), within Chinese history, their depiction of dragons were the sole representation of empirical rule, and gave legitimacy the "power granted by heaven." They were seen decorating the imperial insignias, royal and important architecture, and other official attributes. This then led to the East Asian dragon coming to embody the harmony, richness, and the so-called benevolence of the ruling class. This symbolism is associated with the cosmology of Confucianism and Daoism, from which the Chinese imperial system borrowed the concepts and ideology. Following this in his article, "A Cross-Cultural Analysis of Dragon Symbolism in East Asia," Turdaliev (2025) generalizes this discussion for all of East Asia, suggesting that

despite minor differences, the prosperous symbolism of the dragon remains unchanged in China, Korea, and Vietnam.

In the case of Japan, their drgaon myths represent an interesting example of an intermediate tradition. In his recent research titled, *Dragon Images in Japanese Culture: Genesis and Semantics,* Iztova (2024) writes about the origins and symbolism of Japanese dragons and concludes that they represent a clear derivation of Chinese dragons, that were modified and adapted in way to fit local cultural circumstances. For example, the association between Japanese dragons and violent storm phenomena, dangerous waters, and the unpredictable power of nature is much more prevalent when compared to Chinese dragons. Due to this, Iztova argues it is safe to assume that this difference in the representations reflects the unique environmental context of Japan, namely its geographical exposure to typhoons and tsunamis, as well as the unique theological systems of Shinto and Japanese Buddhism. Finally, in his article, *Slaying the Serpent: Comparative Mythological Perspectives on Susanoo's Dragon Fight,* Weiss (2018) provides a comprehensive comparative analysis of the stories of the battle of the god, Susanoo, against the dragon, and explains how this mythological complex is linked to broader Indo-European and Eurasian myths of storm deities adapted to the unique cosmology of Japan.

The symbolic model also provides insight into the meanings of dragons in the context of religious architecture and religious practices. According to "The meaning of snakes and dragons across world cultures on religious buildings in their iconography," Hoven (2019), snake and dragon iconography have been widely used throughout various religious architectural sites in diverse societies, including ancient Mesopotamian societies and Southeast Asian societies. These beings have always had the same spatial positioning within religious architectural structures and rituals as they are positioned on thresholds or places where humans transition between different worlds or states. This structural consistency in the use of dragons within culturally independent societies implies that the symbolic value of dragons has deep roots in human psychology that can be related to innate cognitive tendencies toward categorization, as described by psychologists like Doubourg and Baumard (2022 in *Imaginary Worlds Through The Evolutionary Lens: Ultimate Functions, Proximate Mechanisms, Cultural Distribution*.

As written in *The Origin of the Dragon,* Rush (2021), dragon mythology can be understood in terms of the integration of both psychological and symbolic theories, as he asserts that the creation of dragon mythologies can emerge from altered states of consciousness and

visions which result from the universal characteristics of the human brain. Although this theory is rather speculative and has not been agreed upon universally among academics, it does reflect the same understanding that underpins the symbolic framework as a whole, namely, that the dragon mythology is a culturally constructed phenomenon that reflects universal human experiences.

In her study, *The Meaning of the Snake in the Ancient Greek World,* Pérez (2020) discusses the symbolism of snakes and dragons in the art of ancient Greece. On the one hand, these images functioned both as a signifier of the ability to heal and of wisdom, while on the other, signifying danger and opposition, which Pérez argues is inherent to beings that stand at the borders of cultural classifications. It should be noted that, in the case under discussion, such a duality is not contradictory, but rather defines the nature of dragons/snakes as symbolic beings. All of these different dragons portrayed an importance to their cultures of origin, which then can help explain how the myths maintained their importance to be culturally maintained, and then later on go to be shared and experienced by other cultures through cultural diffusion.

Cultural Diffusionist Approach

The fourth and final framework of cultural diffusion provides an alternative explanation of cross-cultural similarities found in the myth of dragons. It differs from the previous explanations so far, as it relies on the dynamic interaction between cultures at the social and material level, rather than at the levels of psychological structures or ecology. Although the diffusionist model does not reject the importance of cognitive or environmental factors in the formation of mythologies, it focuses on the fact that, once created within one culture, a mythology can be transmitted to other cultures via trade routes, where it can be adapted into the new culture in unique ways.

The strongest case for diffusionism can be made in those parts of the world where there is clear proof of contact between cultures. One of the best cases for the diffusion of dragon myths is the relationship of the Chinese and Japanese dragons. According to Iztova (2024), the similarity between the dragons of Japan and China shows clearly that the images of the Japanese dragons are borrowed directly from Chinese dragons, with these dragons being transmitted to the Japanese via Buddhism and other cultural contact between the two societies beginning in or around the 6th century. The argument is, that aquatic, serpent-like nature and relation to weather

are not independently developed in the two regions but result from intentional cultural borrowing for use in the appropriate context of each region.

However, another example of the way in which diffusion works can be found in the Mediterranean region. As noted by Núñez, Blanco, and García (2015), there is considerable similarity in the serpentine mythology found in the Near East, North Africa, Egypt, and Southern Europe, which is reflective of, not only, shared environmental settings, but also of the extensive trade, migration, and cultural interactions that have characterized the Mediterranean since the Bronze Age. The traditions of sea-serpent myths in the Ancient Greeks, Phoenicians, and Egyptians can demonstrate enough iconographical and structural similarity to likely imply influence from one another, despite the differences in the specific development of their mythologies. This point is further reinforced by the work of Ogden (2013), in *Drakon: Dragon Myth & Serpent Cult in the Greek & Roman Worlds*, wherein his examination of serpent mythology in the ancient Mediterranean world. Specifically, Ogden notes that the Greek mythology of the drakon included aspects borrowed from Near Eastern traditions while maintaining a unique blend of Greek and Asian draconic traits (Ogden 2013).

In chapter 5 of his book, *The Dragon and the Rainbow,* Robert Blust specifically addresses the ethnology of dragons (77-129, 2023). His writing offers a detailed analysis of dragons within the Austronesian world, including their presence in mythology throughout the Pacific and Southeast Asian archipelagos. This analysis highlights the ability of a primary mythological theme to travel across great geographical distances as it follows the movements of migrating groups of people, such as the expansive seafaring migration of the Austronesians through the Pacific Ocean, with the changes in its depiction accompanying them along the way (Blust 2023). The similarities among the rainbow serpent, the sea dragon, and other island dragons analyzed by Blust are clear enough to indicate an origin in Austronesian mythology, yet distinct enough to indicate the shaping influence of local cultures on received mythology.

The same kind of diffusionist analysis can be found in *On the Trial of Quetzalcoatl/Kukulcan: Tracing Mythic Interaction Routes and Networks in the Maya Lowlands,* by Folan, Bolles, & Ek (2016), where they trace the dissemination of the Quetzalcoatl/Kukulcan mythological figure in the Maya Lowlands. The diachronic changes that take place in relation to this mythic figure's iconography, narrative, and ritualistic role give an example of how the

diffusion process works on a microhistorical scale, showing how this mythical figure could accrue diverse meanings depending on its passage through various groups' hands.

One vital yet commonly overlooked aspect of the diffusionist theory framework is language translation. and how it influences the seeming pervasiveness of dragons across cultures. In *An Analysis of Communicative Translation & Semantic Translation of Peter Newmark*, Luo (2022), discusses the difference between semantic and communicative approaches in translating from one culture's language to another. The first approach, which he called the semantic approach, is a way of translation that is concerned about preserving the syntax and lexicon of the language, thus producing a translation that is accurate but difficult for the recipient language speakers to understand. On the other hand, the second approach he called the communicative method, approaches translation in a way that focuses on conveying the whole meaning of the message in a way that is understandable to the speakers of the other language.

This makes it clear how this definition plays out in relation to dragon mythology. For those early missionaries, travelers, and scholars who encountered mythological animals within foreign cultures, their original way of communicating was very likely communicative translation, where they translated the mythological animal within terms that were closest to those they already had within their cultural vocabulary. As such, a large number of animals that are morphologically and semantically different, such as the Chinese loong, the Mesoamerican feathered serpent, the Japanese ryū, the Mediterranean sea serpent, all became designated as dragons within Western literature, despite being fundamentally different in terms of cultural significance. It must also be pointed out that the meaning of these animals is often very different from their communicative counterparts: much like the way in which the Mandarin language's kanjis for penguin, - 企鹅 - when broken down into it's components - 企 , 鹅 - semantically then translates to "business goose," in english. As such, calling mythological animals, dragons, is more the result of the translator's understanding of the animal and less about its morphological and symbolic equivalence to the concept of a dragon.

It should be pointed out, however, that this linguistic aspect does not negate other explanations; rather, it enhances them. While the cognitive and environmental conditions responsible for the creation of draconic symbolism across disparate cultural contexts are valid, the extent to which disparate cultural traditions tend toward uniformity in their interpretation of the creature can be attributed to translation, which superimposes homogeneity on heterogeneity.

That is to say, it is likely that when colonists, merchants, or travelers arrived in a new place, and were starting the process of translating languages, they were likely taking the ideas of what was being said into account, rather than the original syntactic structure of how the words were being said, and their deeper cultural meaning.

Final Analysis

After exploring these four major frameworks, I have concluded that the answer to my directing questions is not one specific framework, but an integrated approach. After this research, I feel comfortable claiming that none of the frameworks can stand alone, as they are incredibly interconnected, and all become far more powerful when combined together.

The cognitive Framework can account for why dragon myths are created in the first place; why people from widely varying, distinct cultures have a tendency to create mythologies about massive, dangerous and powerful reptilian entities. The theory of the predator template created by Jones (2002), along with evidence from Kawai and He (2016) regarding snake perception, and an overall evolutionary explanation provided by Doubourd and Baumard (2022), all serve as the psychological foundation needed to account for such behavior: an evolutionarily based cognitive system which predisposes people to pay attention, remember and make cultural stories from the type of stimulus dragons usually embody.

The ecological framework offers the needed specificity concerning the environmental inputs involved. In any given culture, flora, fauna, and fossils all contribute to shaping the specific morphology and behaviors of the draconic entities within it. The long-necked aquatic fossils of southern China, the tempests of the Mediterranean, the bird-heavy canopy of Mesoamerica, and the abundance of whales in the northern oceans contribute their share of perceptual and mental stimuli that are then filtered by the psychological base to give rise to culturally specific dragons. It is from the ecological context that one gets to understand the gracileness of Chinese dragons, the combination of bird and reptile elements in the Mesoamerican entities, and the association of chaos with Mediterranean sea monsters.

The symbolic framework helps us understand why dragon myths, after they have been created, become so important from a cultural point of view that it becomes significant enough to be spread, transmitted, and integrated. In other words, an ecological or cognitive imagination producing a vivid picture of a threatening beast alone is not enough for the creation of a

culturally significant myth; rather, the spread will happen only if such an image is assimilated into the system of symbols in which the society structures its perception of power, authority, morality, and cosmic order. Dragons became culturally significant in Western Europe as they fulfilled particular ideological requirements of medieval Christianity. They became culturally significant in imperial China as they fulfilled certain legitimizing functions. They became culturally significant in the Mediterranean to explain the harsh and often dangerous water conditions. They became culturally significant in Mesoamerica because of the religious and agricultural necessities of societies that depend on rainfall.

Lastly, the cultural diffusion framework provides an explanation for the reasons why dragon mythology, despite being independently conceptualized and developed ecologically, displays cross-cultural similarities beyond those which can be explained by independent development alone. This is due to the complex trade and cultural interactions within the Mediterranean, the long history of interaction between China and Japan, and the expansion of the Austronesian culture across the seas, thus creating pathways for the dissemination of mythological elements, which in turn allowed draconic symbolism to migrate between different cultures and be incorporated into the native symbolic systems of the latter. As well, the linguistic translation methods provide the overall explanation for why such morphologically varied creatures all fall under the same mythological phylum. Of course, it does not mean that the cross-cultural pattern is any less valid for being so, but rather points to its true nature.

When all four approaches are considered collectively, they form a holistic explanation for the myths of dragons. Evolutionary psychology offers the cognitive foundation. Local ecology offers the particular environmental triggers that mold morphological features. Cultural symbolism offers the sociological function that guarantees mythic development and continuity, and diffusionism, along with its linguistic component, offers the historical process through which cultural influences affect one another at a regional level. From an anthropological perspective, the importance of this integration lies in its demonstration of how universally held cultural beliefs can develop and persist by the interplay between the universal cognitive processes and particular cultural/environmental processes. While focused on the myths of dragons, these same ideas have the potential to apply equally well to any other universal mythological creature.

Conclusion

In this literature review, four major theoretical perspectives have been analyzed in relation to their explanations regarding the cross-cultural presence of dragons. These include the cognitive evolutionary theory, the ecological theory, the symbolic theory, and the cultural diffusionism theory, with the latter including the linguistic translation element. Each theoretical perspective provides a different explanation related to the same issue and operates on a different analytical level. The cognitive theory explains our universal propensity towards the image of dragons; the ecological theory accounts for culturally-specific manifestations of that image; the symbolic theory explains why dragon myths were such a crucial component of their culture of origin; and the diffusionist theory explains the extra similarity due to cross-cultural exchanges.

The core claim of this review is that the worldwide presence of dragon myths should not be seen as the product of only one of the aforementioned frameworks, but of a conglomeration of the four. This multilevel approach has consequences for anthropology's approach to universals beyond this specific example. Anthropological theories should avoid treating a universal as the consequence of only one framework. Rather, it would be more productive for anthropologists to focus on the circumstances under which certain frameworks take the lead, and others become auxiliary. For instance, in some situations like the case of the dragons in East Asia, diffusion plays the leading role between what is now modern-day China, Japan, Korea, and Vietnam, whilst this approach is not able to explain East Asian dragons versus other myths throughout the world.

The anthropological relevance of such an observation is further expanded by applying it to the second research question posed at the beginning of this review: What does it tell us about the human relationship with the world? The dragon myth, with its universal presence in human cultures and its variation from culture to culture, is one such example of how humans utilize imagination and symbolism in understanding their surroundings, the threats they face, and the social systems they create and perpetuate - a trait that can be argued to be what makes us human, and separates us from other beings. It is not surprising that the same cognitive processes that generated the menacing wyrm of medieval Europe also generated the auspicious loong of imperial China. This is because of the ability of human symbolic systems to be highly malleable, capable of transforming the same underlying psychological framework in light of different ecological, historical, and social dynamics into culturally unique yet recognizably similar representations.

Annotated Bibliography

Blust, Robert. 2023. "Chapter 5: The Ethnology of the Dragon." In *The Dragon and the Rainbow*, 77–129. Leiden, The Netherlands: Brill.

Blust's chapter presents an in-depth ethnological analysis of mythological figures resembling dragons in the Austronesian-speaking world, showing how the mythological complex has traveled and been reworked by seafaring communities over millennia throughout the Pacific and Southeast Asia. Using language, comparative folklore, and material artifacts, Blust is able to illustrate how mythical creatures like the rainbow serpent and sea dragon contain similar structures, indicative of shared origin within Austronesian culture, yet differ significantly in their localized versions. The use of this source in the paper helps support the application of the diffusionist approach to non-Mediterranean, non-East Asian transmission processes. Blust's analysis reveals that mythical imagery diffused through migration routes rather than through previously established trade routes. It is especially helpful for the paper because it suggests that diffusion is not at odds with local adaptation and that recipient societies have actively transformed the mythology they receive.

Doubourg, Edgar, and Nicolas Baumard. 2022. "Imaginary Worlds Through the Evolutionary Lens: Ultimate Functions, Proximate Mechanisms, Cultural Distribution." *Behavioral and Brain Sciences* 45.

The analysis of Doubourd and Baumard situates the investigation of mythology and fantasy within an extensive evolutionary context, arguing that the imaginary world's content is not random but reflects evolved psychological mechanisms that respond to adaptive issues such as predation, hierarchical social structures, and environmental unpredictability. This research outlines the evolutionary origins of a psychological disposition from its cultural manifestation. It elaborates on Jones' (2002) concept by situating the predator template theory within the broader context of evolutionary cognition’s impact on human culture and imagination. This source is important for anchoring the cognitive theoretical model in the scientific literature on human behavior rather than relying on Jones's highly speculative book. It also helps explain why dragons,

among many other mythological creatures, feature prominently in human cultures' imaginations.

Folan, William J., David D. Bolles, and Jerald D. Ek. 2016. "On the Trail of Quetzalcoatl/Kukulcan: Tracing Mythic Interaction Routes and Networks in the Maya Lowlands." *Ancient Mesoamerica* 21 (2): 293–318.

The article explores the geographical distribution and development of the tradition of Quetzalcoatl/Kukulcan throughout the Maya lowlands, using archaeological data, iconography, and trade/political network mapping to outline how the draconic deity migrated across cultures within that region. Through its use, the authors illustrate how, even in a region that should be culturally connected, a single mythological entity may accrue many differences in meaning and morphology as it travels from culture to culture. In this paper, the source serves two roles: first, as the primary example of a Mesoamerican diffusionist theory in action, given the significant archaeological evidence of this phenomenon in the region; second, as proof that the Quetzalcoatl myth emerged completely separately from any dragon-related myths in Europe or East Asia.

Hoven, Stijn Van Den. 2019. "The Meaning of Snakes and Dragons Across World Cultures on Religious Buildings and in Their Iconography." *Academia*.

This paper examines the role of snakes and dragons in the construction of sacred architecture across the world, in a range of cultures, from the ancient civilizations of Mesopotamia and Egypt to Southeast Asian and pre-Columbian Mesoamerican peoples. The author's focus is on the figure's positioning within the structure, both physically and symbolically, and on how it informs an understanding of the figure's meaning as a whole. Hoven argues that the prevalence of this imagery worldwide results from the serpent and dragon's roles as liminal figures that mediate between humans and deities, as well as between order and chaos. In the current paper, the text cited from this source is used to show that the geographic and historical reach of serpent imagery cannot be explained purely by processes of cultural diffusion, and also to illustrate the symbolic argument regarding the positioning of the figures.

Iztova, N. N.. 2024 *Dragon Images in Japanese Culture: Genesis and Semantics*. Volume 7 (issue 2) Russian Japanology Review: 94-114 https://www.japanreview.ru/jour/article/view/109/85

Iztova explores the origins and symbolic functions of dragon imagery in *Japanese* mythology, focusing on how Japanese dragons developed through a combination of indigenous beliefs, Shinto religious ideology, imported Chinese influences, and regional environmental disasters, such as severe storms and tsunamis. The article draws on linguistic evidence, classical Japanese texts, and religious iconography to show how dragons in Japan operate as protective, ambivalent, or liminal figures, rather than purely malevolent or purely gracious creatures. This source is perfect for my personal argument that the development of dragons is not just individual but also not entirely interconnected, because it shows how dragons originated in Japan while adopting attributes of their Chinese cousins. It also helps show the comparative differences between East Asian and Western/Near Eastern traditions.

Jones, David E. 2002 *An Instinct for Dragons*. New York: Routledge.

The purpose of the inclusion of *An Instinct for Dragons* is that it is the most influential anthropological explanation for the global recurrence of Dragon Myths. David E. Jones, an anthropologist, draws on evolutionary psychology as well and argues that humans possess inherited predator-detection instincts shaped by early evolutionary threats, such as snakes, raptors, and large carnivorous mammals. According to him, dragons are a composite projection of these instincts, which explains why cultures with no contact with one another still developed similar monstrous creatures. The book is one of the few that combines folklore with psychological theory. This source is valuable because it provides a clear, well-articulated psychological/cognitive explanation and establishes a framework for understanding dragons as universal archetypes rather than as products of cultural diffusion.

Kawai, Nobuki, and Hongshen He. 2016. "Breaking Snake Camouflage: Humans Detect Snakes More Accurately than Other Animals Under Less Discernible Visual Conditions." *PLoS ONE* 11 (10).

This experiment illustrates how snakes that are visually camouflaged can be detected faster and with greater accuracy in obscured/blured visual conditions by humans compared to other species, thus lending empirical psychological evidence to the "snake detection hypothesis," which claims that the human visual system has been evolutionarily selected specifically due to the necessity of being able to detect snakes within their habitat. In the paper, the source serves as primary scientific evidence validating Jones' (2002) hypothesis on the predator template. This is done to show that there is actual perceptual significance in serpent imagery that can explain why it was so prevalent across different world mythological traditions.

Khalifa-Gueta, Sharon. 2001 *The Evolution of the Western Dragon*. Volume 4 (Issue 4) Athens Journal of Mediterranean Studies: 265-290.

This article examines the development of dragon imagery in the Mediterranean and Western European worlds, tracing how simple serpent-like figures gradually evolved into the complex winged fire-breathers we typically think of today. Using textual analysis and iconography, the author argues that Western Dragons emerged from the interplay of natural predators, Near-Eastern mythological imports, and later Christian reinterpretations. The article provides a well-supported historical narrative that puts dragons within shifting cultural and religious contexts and highlights the changes over time. It is especially useful because it grounds Western dragon myths in concrete historical contexts, demonstrating how environments, cross-cultural exchange, and religious/ideological frameworks shape the meaning of dragons within a region

Lorenzen, Michael. 2025. "Dragons in Mythology: A Literature Review." *ResearchGate*.

Lorenzen explores in his literature review the various academic perspectives on dragon mythologies throughout the world, combining scholarship from such disciplines as anthropology, folkloristics, paleontology, and cognitive science to provide a current status of the topic. One of the author's contributions to the existing knowledge about dragons is by reviewing literature that examines how dragon myths correlate geographically with the presence of accessible fossil records of dinosaurs and large reptiles; as noted by the author, it seems as if draconic beliefs in different societies tend to

emerge more often in places where there is an abundance of fossils. This source is cited in the ecological perspective section to create an academic background for the fossil encounter component.

Luo, Ni. 2022. "An Analysis of Communicative Translation and Semantic Translation of Peter Newmark." *International Journal of Social Science and Education Research* 5 (10): 63–70.

The Luo piece explores Peter Newmark's taxonomy of semantic and communicative translations within cross-cultural linguistic communication. In semantic translations, the focus is on preserving syntax and lexicon in the original language, resulting in exact yet possibly unclear translations. Communicative translations emphasize the meaning of the whole rather than specific aspects, and the translated material reflects native usage patterns, though some clarity may be lost in conveying the original message. The article draws on examples from Chinese-English translations to demonstrate the marked difference between the two translation strategies. Within the Luo piece, this literature becomes the basis of the linguistic element within the diffusionist framework, using the terminology made to support the claim that the cross-cultural universality of the dragon myth, in part, owes to the practice of communicative translations, where the differences in morphology and symbolic meaning among local mythological figures were translated into English using the term "dragon," due to its availability within the translators' culture.

Núñez, Eloy M., Blanco, Antonio C., and García, Aitana M. 2015 *The Serpent as a Pan-Mediterranean Myth.* Volume 6 (Issue 4) Mediterranean Journal of Social Sciences.

This article is a comparative analysis of serpent and serpent-dragon myths across the Mediterranean, arguing that these symbols form a shared 'cultural vocabulary' linking Greek, Near Eastern, North African, and Southern European Societies. The authors explore recurring themes, such as chaos monsters, sea serpents, and underworld guardians, using literary analysis, archeological references, and regional folklore as evidence. Their approach focuses on similarities in environmental conditions, trade networks, and cross-cultural contact, which aided in the spread and transformation of the

serpent-based mythological figures. It's valuable because it strengthens the diffusionist and cultural-contact side of the argument, illustrating how the broad strokes of myths move, adapt, and meld across connected cultural zones.

Ogden, Daniel. 2013. *Drakon: Dragon Myth and Serpent Cult in the Greek and Roman Worlds*. Oxford: Oxford University Press.

Ogden's work is a monographic overview of the significance of the drakon or the serpent and the dragon in the myths, religions, and cults of the Greeks and Romans, analyzing the various roles played by these creatures as protective deities, monstrous opponents, underworld-dwelling entities, and beings that are revered as part of the actual practice of Greco-Roman religions. Analyzing sources such as literature, inscriptions, and archaeological finds, Ogden discusses the development of serpent mythology in ancient Greece and Rome and how it incorporated Near Eastern mythological material into a distinctive Greco-Roman context. In the paper, this source is used in the background section to provide a starting point for comparison with other cultures, and in the diffusionist section to support the claim that the Greco-Roman drakon mythology integrated Near Eastern material through Mediterranean culture.

Pérez, Diana R. 2020. "The Meaning of the Snake in the Ancient Greek World." *Arts* 10 (1): 2.

Pérez writes about how the snake was a symbol of healing, wisdom, prophetic knowledge, death, and danger in Ancient Greek material culture, including ceramics, sculptures, architecture, and votive offerings. She claims that the ambivalent nature of such symbols does not reflect a logical inconsistency but rather is an inherent quality of animals that occupy liminal roles in cultural classification - animals can possess opposite characteristics at once and thus be employed in various rituals, narratives, and politics. In the paper, the source is used in the symbolic framework section as an example of the inner complexity of draconic symbolism within a given culture, thereby supporting the thesis that social structure influences the symbolic function of dragon-like images. It is an interesting example that contradicts simple oppositions of symbols of the East and the West, as it proves that even within Western tradition, a snake image had multiple, opposing meanings.

Rose, Carol. 2001. *Giants, Monsters, and Dragons: An Encyclopedia of Folklore, Legend, and Myth*. New York: W.W. Norton.

This work by Rose offers an encyclopedia reference for dragons, monsters, and giant creatures from the realm of folklore, legends, and myths all over the globe, offering standardized information on numerous varieties of dragons and giant creatures from all over the globe. As a reference book, it does not offer a theoretical point for discussion; however, it provides an empirical review of the presence of draconic creatures across various civilizations worldwide. This source supports the introductory statement in the paper that numerous societies have created its own type of large reptilian mythological figure. It helps support the credibility of the information about various draconic creatures that cannot be found in the four main cultural regions discussed in the paper.

Rush, John A. 2021. "The Origin of the Dragon." *Anthropology of Consciousness* 31 (2): 153–173.

Rush hypothesizes that the source of the dragon myth lies in experiences of altered states of consciousness and visions. According to him, visual imagery during a psychedelic or trance state triggers the creation of serpent and dragon images due to common neurologic processes within the human brain. For his thesis, Rush employs anthropological data of the use of psychoactive drugs in a shamanic or spiritual context, as well as data from modern neuroscience about altered state experiences. Although this hypothesis is highly speculative and controversial, Rush's article is helpful to the paper for two reasons: it provides an example for the ecological framework that stresses the importance of experiencing mysterious objects in the environment as the base for myths creation; and it shows that the human brain has a tendency towards reptiles and other predators, which goes towards the cognitive framework and strengthens Jones' (2002) and Doubourg and Baumard's (2022) arguements.

Spikeman, Stephan N.F., Wei Wang, Lijun Zhao, Oliver Rieppel, Nicholas C. Fraser, and Chun Li. 2024. "*Dinocephalosaurus orientalis* Li, 2003: A Remarkable Marine

Archosauromorph from the Middle Triassic of Southwestern China." *Earth and Environmental Science Transactions of the Royal Society of Edinburgh* 114 (3–4).

In this paleontological study, the anatomy of *Dinocephalosaurus orientalis*, a Middle Triassic marine archosauromorph that lived off the coast of modern-day southern China, is described in detail. This creature, known for its exceptionally long neck, snake-like body, and overall anatomy that closely resembles the iconography of dragons depicted in China, is discussed within the broader context of the evolution of marine reptiles during the Triassic period and of archosauromorphs. This source is used to provide an example of the fossil encounter scenario in the context of dragon mythology, within the ecological framework. Although the authors do not explicitly claim a cause-and-effect link between Dinocephalosaurus and dragon myths, the occurrence of fossils like these in the same geographical area with a rich dragon mythology offers a plausible explanation that encounters with fossils of such creatures may have played an important role in creating dragon myths.

Turdaliev, Olimjon D. 2025. "A Cross-Cultural Analysis of Dragon Symbolism in East Asia." *International Journal of Literature and Languages* 5 (3): 107–109.

Turdaliev investigates the use of dragon imagery across the East Asian cultural region, encompassing China, Korea, Vietnam, and Japan. According to Turdaliev, although dragons vary across East Asian cultures, there is a fundamental association cluster that includes positive connotations such as imperial sovereignty, celestial power, water, and luck. It is notable that, unlike the West, where dragons have negative connotations, this cluster of positive associations is shared throughout East Asia. Therefore, Turdaliev uses comparative cultural studies to explore the cross-cultural spread of the cluster across East Asia through centuries of cultural exchange. The source is used within the paper's symbolic framework, where Zhang's (2024) distinction between China and the West is extended to East Asia. The extension shows that the positive nature of dragons is not an anomaly found only in China, but a common phenomenon across East Asian cultures bound by a historical exchange channel, which also then strengthens my argument of the multi-framework approach to mythological studies, where cultural diffusion happens in pockets throughout the world, but not necessarily as a whole.

Weiss, David. 2018. "Slaying the Serpent: Comparative Mythological Perspectives on Susanoo's Dragon Fight." *Journal of Asian Humanities at Kyushu University* 3: 1–20.

Weiss writes a comparative mythological study of the story of Susanoo and his defeat of the eight-headed dragon, Yamata no Orochi, which features prominently in Japanese mythology as part of the myths surrounding the storm god Susanoo. The author places the myth within the broader context of East Asian dragon lore and the pan-Eurasian category of storm god legends in Indo-European cultures. Weiss shows how the tale of Susanoo incorporates elements from the iconography and mythology of the Chinese dragon tradition while also transforming them in accordance with Japanese cosmology and theology associated with Shintoism and early Buddhism in Japan. In the paper, the use of this text helps explain Japan's intermediate position in its relationship to China and Western dragons by offering a specific example of a Japanese myth that draws on imported mythology to create a Japanese narrative.

Zhang, Le. 2024 *Different Symbolic Meanings between Chinese Dragon 'Loong' and Western Dragon*. Volume 6 (Issue 8) Region-Educational Research and Reviews: 147-290

Zhang compares the Chinese Dragons with their Western-European counterparts, arguing that they share superficial similarities but differ significantly in their symbolic meanings, cultural roles, and historical developments. He uses semiotics and literary analysis for his approach. The article contrasts Chinese dragons, symbols of harmony, imperial authority, water, and prosperity, with Western dragons, typically portrayed as destructive, chaotic, or even demonic. Zhang demonstrates how environmental factors, religious structures, and political structures shaped each tradition's dragon mythology differently. This source is *critical* for highlighting the cultural specificity viewpoint, due to it challenging assumptions that all 'dragons' are the same, and helps address the problem of definition and variation across cultures.